Dataset Academic Summary 01 — MLAAD

MLAAD: The Multi-Language Audio Anti-Spoofing Dataset

Müller et al., IJCNN 2024; arXiv v11, 2026

The paper *MLAAD: The Multi-Language Audio Anti-Spoofing Dataset* by Müller and colleagues presents a multilingual synthetic-speech dataset, intended to support the development and evaluation of voice forgery detection systems. The study focuses on the generalization problem of forgery detectors to data sources and synthesis systems not included in training, while emphasizing the linguistic bias characterizing many Anti-Spoofing datasets, which focus mainly on English and Chinese.

Unlike studies proposing new detection architectures, the central contribution of MLAAD is in the area of building the training data. The researchers seek to examine whether using a dataset containing a wide variety of languages and Text-to-Speech systems may lead to learning forgery features with a broader generalization ability.

The dataset is based on the M-AILABS Speech Dataset, which includes recordings of human speech. The researchers used the texts of the source recordings, and sometimes their translation into additional languages, in order to produce synthetic recordings by means of diverse TTS systems. In models supporting this, voice-reference recordings were also used. Alongside the audio files, Metadata is kept describing, among other things, the recording's language, the TTS model, the architecture, the text, the source recording, and sometimes also the reference speaker. This information makes it possible to analyze the source of the examples and the possible associations between them.

MLAAD has expanded across several versions since its original publication. Version 11, described in the version of the paper read, includes 1,153.6 hours of synthetic speech in 54 languages, produced by means of 205 TTS models belonging to 127 architectures. However, it is important to distinguish between the development of the dataset and the scope of the reported experiments: the central Cross-Dataset evaluation was carried out by means of MLAAD v1 only, which includes 63 hours, 21 TTS models, and 22 languages. The authors note that the experiments were not rerun for the later versions, and therefore the expansion of the dataset should not be seen as empirical proof of a further improvement in generalization ability.

For the purpose of evaluating MLAAD, the researchers trained three voice forgery detection systems — RawGAT-ST, SSL-W2V2, and WhisperDF — on several different training datasets, including MLAAD v1, ASVspoof 2019, FakeOrReal, and InTheWild. Subsequently the models were tested on additional evaluation datasets under Cross-Dataset conditions. The training included class balancing by means of Random Undersampling and data augmentation by means of noise, music, and audio coding. The metric used by the researchers was Accuracy, among other things because some of the evaluation sets contained fake examples only and therefore did not allow computing EER.

The results demonstrate that the choice of training dataset significantly affects the generalization ability of the detectors. In particular, MLAAD and ASVspoof 2019 showed complementary advantages: each of the training sources led in a different part of the evaluation scenarios. For example, for WhisperDF on WaveFake, a model trained on MLAAD v1 achieved a mean Accuracy of 97.2% compared with only 2.2% for a model trained on ASVspoof 2019. Conversely, on the ASVspoof 2021 DF dataset, a better result was obtained for SSL-W2V2 trained on ASVspoof 2019 than for the same type of model trained on MLAAD v1.

These findings support the conclusion that different training sources may confer different advantages under changing evaluation conditions. However, they do not provide proof that combined training on both datasets would necessarily be better than separate training, or that the diversity of languages is the direct cause of the improvement. The number of languages, the amount of data, the diversity of TTS systems, and the features of the original recordings change together, and therefore their relative contribution is not fully isolated. In addition, some of the results are characterized by high variance between runs, and an Accuracy metric on a single-class dataset does not make it possible to draw conclusions about the overall discriminative ability between real and fake speech.

For the present study, which deals with voice forgery detection and generalization to unknown attacks, MLAAD is especially relevant as a candidate for expanding the training data. The dataset may add a diversity of synthesis systems beyond the systems represented in ASVspoof 5 and in CodecFake. Alongside this, its use requires a careful check of Data Leakage and Shortcut Learning risks, since MLAAD is based on M-AILABS, from which some of the real recordings in the course data also come. Possible overlaps in source recordings, speakers, texts, or recording features may lead to an optimistic evaluation that does not reflect true generalization ability.

Accordingly, it is proposed to first examine a controlled subset of MLAAD, while checking Metadata, licensing, source associations, and the feasibility of splits separating speakers and related recordings. Subsequently it will be possible to compare a model trained on the course data alone to the same model after expanding training by means of MLAAD, and also to examine its contribution relative to PhonemeDF and to the combination of both datasets. In order to distinguish between the effect of the amount of data and the effect of the diversity of forgery systems, controlled experiments in which the example budget is kept similar will also be required.

The central research conclusion is that the composition of the training data constitutes a significant factor in the generalization potential of voice forgery detection systems, but the effect of the diversity of languages, generation systems, and recording sources requires controlled evaluation and cannot be inferred from overall classification performance alone.

Reference

Müller, N. M., Kawa, P., Choong, W. H., Casanova, E., Gölge, E., Müller, T., Syga, P., Sperl, P., & Böttinger, K. (2024). *MLAAD: The Multi-Language Audio Anti-Spoofing Dataset*. International Joint Conference on Neural Networks (IJCNN). https://arxiv.org/abs/2401.09512

Source note: the summary refers to arXiv version v11 from August 2026. The reported generalization results are based on MLAAD v1. The recommendations for using external data within the course project are our own planning conclusions and not part of the original experiments.